

Part 1 Design 和 Requirement 相关
1.Interaction Design

Definition: Designing interactive products to support the way people **communicate** and **interact** in their everyday and working lives. **Interaction design is the umbrella term** covering all of these aspects: Fundamental to all disciplines, fields, and approaches concerned with researching and designing computer-based systems for people.

Core characteristics: 1.Users should be involved throughout the development of the project 2.Specific usability and user experience goals need to be identified, clearly documented, and agreed to at the beginning of the project 3.Iteration is needed through the core activities

Importance of involving users:

- 1.Understanding of users' goals leading to better products.
- 2.Expectation management
- 3.Ownership

Degrees of user involvement:

- (1)Member of the design team:
Full-time: constant input, but lose touch with users 可能逐渐脱离普通用户群体
Part-time: patchy input, and very stressful 参与可能不稳定, 也有压力
Participatory design: involve all stakeholders in the early stages of design
- (2)Face-to-face group or individual activities
- (3)Online contributions from thousands of users
→ Online Feedback Exchange (OFE) systems
→ Crowdsourcing design ideas
→ Citizen science
- (4)User involvement after product release
Customer review analysis

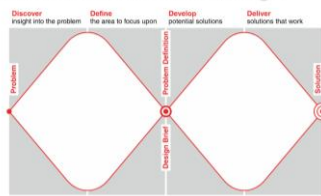
Usability and User experience

Usability(能用,Objective): how useful or productive a system is from its own perspective. 关键词:effectiveness, efficiency, safety, utility, learnability, memorability
User experience(享用,Subjective): how users experience an interactive product from their own perspective. 关键词:Enjoyable, Motivating, Experiencing flow, Engaging, Satisfying

Four basic activities of Interaction Design:

- 1.Discovering requirements for the interactive product.
- 2.Designing alternatives that meet those requirements.
- 3.Prototyping the alternative designs so that they can be communicated and assessed.
- 4.Evaluating the product and the user experience it offers throughout the process.

The double diamond of design



Practical Issues in Interaction Design

User-centered approach: 3 principles

- 1. **Early focus on users and tasks:** directly studying cognitive, behavioral, anthropomorphic, and attitudinal characteristics
- 2. **Empirical measurement:** users' reactions and performance to scenarios, manuals, simulations, and prototypes are observed, recorded, and analysed
- 3. **Iterative design:** when problems are found in user testing, fix them and carry out more tests

Early focus on users and tasks

- 1. Users' **tasks and goals** are the driving force behind the development.
- 2. Users' **behaviour and context of use** are studied, and the system is designed to support them.
- 3. Users' **characteristics** are captured and designed for.
- 4. Users are **consulted** throughout development from earliest phases to the latest.
- 5. All design decisions are taken within the context of the **users, their activities, and their environment**

Who are the users/stakeholders?

User: are not very obvious, many products are intended for use by large sections of the population, so user is "everybody".More targeted products are associated with specific roles

Stakeholders: The individuals or groups that can influence or be influenced by the success or failure of a project. Larger than the group of direct users. Identifying stakeholders helps identify groups to include in interaction design activities

What are the users' needs?

Users rarely know what is possible: "undreamed-of" needs.
Instead: Explore the problem space, Investigate who are the users, Investigate user activities to see what can be improved, Try out ideas with potential users.
Focus on peoples' goals, usability, and user experience goals, rather than expect

stakeholders to articulate requirements.

2.Requirements

Definition:A requirement is a statement about an intended product that **specifies what it is expected to do or how it will perform**.
Different Forms: Atomic requirement shell : 结构化写需求的方法
User stories: As a <role>, I want <behavior> so that <benefit>

Six most common types of requirements

- 1.**Functional requirements:** What the system should do
- Non-functional requirements:** The characteristics (sometimes called constraints) of the product
- 2.**Data requirements:** What kinds of data need to be stored? How will they be stored (for example, database)?
- 3.**Environment requirements or context of use:** → Physical: dusty? noisy? vibration? light? heat? humidity? (for example, in a hospital)
→ Social: collaboration and co-ordination, data sharing, distributed, synchronous or asynchronous, privacy
→ Organizational: user support, communications structure and infrastructure, availability of training
→ Technical: On what technologies will it run or need to be compatible?
- 4.**Users characteristics:**
→ Nationality, educational background, attitude to computers
→ System use: novice, expert, casual, frequent
→ User profile
- 5.**Usability goals**
- 6.**User experience goals**

Data gathering for requirements

- (1)Interviews, observations, and questionnaires
 - (2)Studying documentation(Good for understanding legislation and getting background information Does not involve stakeholder time)
 - (3)Researching similar products(Good for prompting requirements)
- Combining Data Gathering 不同方法最好混合使用
相关工具方法:

Using probes to engage with users 一种用来引导用户记录、表达或反思自己生活和行为的工具。Many types of probes:
→ Designed to prompt users into action
→ For researchers to learn about users
Contextual Inquiry (情境访谈) 到用户真实工作或生活场景中, 边观察边访谈。Part of Contextual Design, but also used on its own to gather requirements.
One-on-one field interviews (contextual interviews)

Contextual Inquiry Four main principles:

Context: Going to the user, wherever they are, and seeing what they do as they do it
Partnership: User and interviewer explore user's life together
Interpretation: Observations interpreted by user and interviewer together
Focus: Project focus to understand what should be paid attention

Contextual inquiry guided by seven "cool concepts" two groups: 1.**Joy of life concepts** (How products make our lives richer and more fulfilling): Accomplish (empower users) , Connection (enhance real relationships), Identity (support users' sense of self) ,Sensation (pleasurable moments) 2.**Joy of use concepts** (Describe impact of using the product) :Direct in action (provide fulfillment of intent),The hassle factor (remove all glitches and inconveniences),The learning delta (reduce the time to learn)

Interview in four parts : Overview ,Transition, Main interview, Wrap-up
Following interview: interpretation session, Contextual design models are created or consolidated, Most relevant models are chosen by team, out of 10 suggested

当我们已经理解了用户、问题空间和需求之后, 需要用 brainstorming 产生更多可能的设计想法。

- 1. Include participants from a **wide range of disciplines**, with a broad range of experience
- 2. Don't ban silly stuff
- 3. Use catalysts for further inspiration
- 4. **Keep records:** Capture every idea, without censoring
- 5. **Sharpen the focus**
- 6. Use warm-up exercises and make the session fun

Persona and Scenario

Persona: Rich descriptions of typical users, not specific people.
→ Capture a set of user characteristics (user profile)
→ Synthesised from real people based on user research
→ Typical, not idealised
→ Bring to life with name, characteristics, goals, and personal background
→ Relevant to product under development
Two goals of persona
→ **Helps designer with design decisions and**
→ **Reminds team about who will use the product**
→ Develop a small set of personas **with one primary**
Scenario: An informal narrative story, simple, 'natural', personal, and not generalizable.

- Textual descriptions
- Animations, audio or video
- Example animation scenarios

Scenario 和 Persona

Persona: 用户是谁 **Goal:** 用户目标是什么
Scenario: 用户在什么情境下完成目标
关系是: Persona 提供人物背景, scenario 描述这个人物如何达成目标。

Use Cases

关注用户和系统之间的交互步骤。
Focus on functional requirements and capture interaction. Can be used in design or to capture requirements. Use cases are step-by-step descriptions of interactions, user stories focuses on outcomes and user goals

Two styles:

- 1.**Essential use cases:** division of tasks, no implementation detail 关注 user intention 和 system responsibility,它不会写太多界面细节。
- 2.**Use case with normal and alternative courses:** more detail

3.Design Principles

Accessibility and Inclusiveness
Core pillars of Accessibility and Inclusiveness: Empath 同理心, Usabilit 可用性, Equit 公平性
Disability in HCI:
The Medical Model: Historically viewed disability strictly as a personal health condition or deficit to be cured by doctors.
The Social Model: It states disability is a mismatched interaction between a person and an unaccommodating environment.
The Designer's Role: By recognizing that poor design creates barriers, technologists can take full responsibility for engineering accessible interfaces.

Universal vs. Inclusive

Universal Design strives to create a single optimal solution that works for everyone uniformly, much like a standard wheelchair ramp at a building entrance.追求一个统一的解决方案, 尽量适用于所有人。
Inclusive Design recognizes that human diversity is too complex for one solution. It focuses on providing a flexible range of ways to participate.人的多样性太复杂, 很难一个方案解决所有问题, 所以应该提供灵活的参与方式。
Accessibility and inclusiveness
Accessibility: the extent to which an interactive product is accessible by as many people as possible 交互产品能被尽可能多的人访问和使用 Focus is on people with disabilities; for instance, those using android OS or apple voiceover
Inclusiveness: making products and services that accommodate the widest possible number of people 范围更广

Impairment can be categorized: Permanent, Temporary, Situational
解决工具: Screen Readers 屏幕阅读器, Braille Displays 盲文显示器, Adaptive Switches 适配性开关

Principles of Inclusive Design

- 1.Recognize Exclusion 2.Solve for One, Extend to Many 3.Learn from Diversity

Design Principles

Definition: Design principles are fundamental, overarching heuristics and guidelines that dictate how a product should be structured, behave, and appear. They are the established rules of engagement for interaction.
Meaning: They act as the critical bridge between human psychology and digital interfaces. By adhering to these principles, we align system mechanics with natural human cognitive models and expectations.
Primary Functions: Guide Decisions, Evaluate Quality, Ensure Standardization
The Design Principles
Visibility: The more visible functions and system status are, the more likely users will be able to know what to do next. Conversely, when functions are hidden, they become significantly harder for users to discover and utilize.
Feedback: Feedback refers to the system sending information to users about the results of their operations.
Constraints: Constraints refer to limiting the possible actions that can be performed to guide users in using a product or system correctly.
Consistency: Consistency refers to the use of similar operating methods and elements to accomplish similar tasks when designing user interfaces. **Internal consistency:** Internal consistency maintaining consistency in the design of operational behavior within the same application. **External consistency:** designing operations, interfaces be the same across different applications and devices.
Affordance: The physical or digital properties of an object that show users what actions are possible. Eg.Signifiers: The specific visual cues that communicate where the action should take place.

Shneiderman's Eight Golden Rules of Interface

Design

Strive for consistency(Workflows/Processes, Functionality, Appearance, Terminology)用 information scent

Enable frequent users to use shortcuts (Keyboard shortcuts, Hidden "power user" features, Automation)允许高频用户使用快捷方式

Offer informative feedback (Relevant, Fits importance and urgency, Comprehensible and meaningful, Within appropriate context (time and place))

Design dialog to yield closure (Grouping of actions, Explicit completion of an action, Well-defined options for the next step)

Offer simple error handling (Error prevention over error correction, Automatic detection of errors, Clear error notifications, Hints for solving the problem)

Permit easy traversal of actions (No interference with workflow, More freedom for the user, Single-action undo vs. action history)

Support internal locus of control (The user commands, the system obeys, Strongly relies on the informative feedback, "The principle of least surprise")

Reduce short-term memory load (Clear structure: windows, dialogs, app in its entirety, "Recognition over recall", Implicit help)

Alternative designs

The Dual Process of Design: Divergent Thinking and Convergent Thinking

Methods for Generating Alternatives Brainstorming, Crazy Eights, Morphological Analysis

Evaluating the Alternatives: Heuristic Evaluation, A/B Testing, Qualitative User Feedback

Questionnaires

Questions can be closed-ended or open-ended. Closed questions are easier to analyze, and may be distributed and analyzed by computer. They can be administered to large populations. Disseminated by paper, email and the web.

Pros:1. You can collect data from a large number of people, at a relatively low cost. 2. You can get an overview of a population of users in a short amount of time. 3. Surveys do not require any special equipment 4. Surveys are generally approved by institutional review boards because they are typically non-intrusive.

Cons:1. Surveys are good at getting shallow data from a large number of people, but are not good at getting "deep" data. 2. Since surveys are usually self-administered, it is usually not possible to ask follow-up questions 3. Surveys can lead to biased data when the questions are related to patterns of usage, or feelings about a previous experience, rather than clear factual phenomena.

Encouraging a good response:1.Make sure that the purpose of study is clear. 2.Promise anonymity. 3.Ensure that questionnaire is well designed. 4.Offer a short version for those who do not have time to complete a long questionnaire 5.Follow-up with emails, phone calls, or letters. 6.Provide an incentive 7.If mailed, include a stamped, addressed envelope

Deploying online questionnaires: 1. Plan the timeline 2. Design the questionnaire offline 3. Program/complete online survey 4. Test the survey to make sure that it behaves as you would expect 5. Test it with a group that will not be part of the survey to check that the questions are clear 6. Recruit participants

Interviews

四种类型 1.Unstructured: Not directed by a script. Rich but not replicable.2.Structured: Tightly scripted, often like a questionnaire. Replicable but may lack richness. 3.Semi-structured: Guided by a script, but interesting issues can be explored in more depth. Can provide a good balance between richness and replicability.4.Focus groups: A group interview
Two types:1.'Closed questions' have a predetermined answer format 2.'Open questions' do not have a predetermined format. Closed questions are easier to analyze
Running the interview:1. Introduction 2. Warm-up 3. Main body 4. A cool-off period 5. Closure

Pros:1.Go deep: encourage reflection and consideration 2.Flexible: open-ended and exploratory

Cons:1.Skill to manage2.Time and resource intensive3.Data analysis4.Recall problems
Five key issues of data gathering:1.Setting goals 2.Identifying participants 3.Relationship with participants 4.Triangulation 5.Pilot studies 预测测试

Heuristic Evaluation

Nielsen's heuristics: 1. Visibility of system status 2.Match between system and real world 3.User control and freedom 4.Consistency and standards 5.Error prevention 6.Recognition rather than recall 7.Flexibility and efficiency of use 8.Aesthetic and minimalist design 9.Help users recognize, diagnose, recover from errors 10. Help and documentation

Pros: 1.Few ethical and practical issues to consider because users are not involved 2.Can be difficult and expensive to find experts 3.Best experts have knowledge of application domain and users
Cons: 1.Important problems may get missed 2.Many trivial problems are often identified, such as false alarms 3.Experts have biases

Part 3 Evaluate 相关

Iterative design and evaluation is a continuous process.

Why: To check users’ requirements and confirm that users can utilize the product and that they like it.

What: A conceptual model, early and subsequent prototypes of a new system, more complete prototypes, and a prototype to compare with competitors’ products.

Where: In natural, in-the-wild, and laboratory settings.

When: Throughout design; finished products can be evaluated to collect information to inform new products.

Types of evaluation methods

- 1. **Controlled settings** that directly involve user
- 2. **Natural settings** involving users. For instance, online communities and products that are used in public places. Often there is little or no control over what users do, especially in in-the-wild settings
- 3. **Any setting** that doesn’t directly involve users. For example, consultants and researchers critique the prototypes, and may predict and model how successful they will be when used by users

Method	Controlled settings	Natural settings	Without users
Observing	✓	✓	
Asking users	✓	✓	
Asking experts		✓	✓
Testing	✓		
Modeling			✓
Method	Controlled settings	Natural settings	Without users
Examples	✓ Usability testing ✓ Experimental design	✓ Field study	✓ Heuristic evaluation ✓ Analytics ✓ A/B testing ✓ Predictive models

Practical challenges of evaluation

- 1.Participants’ consent 参与者知情权
- 2.Interpreting data: Reliability, Validity, Ecological validity, Biases, Scope 解读数据

Usability testing

特点: 1.**Controlled settings** 2.Users are observed and timed 3.Data is recorded on video, and key presses are logged 4.The data is used to calculate performance times and to identify and explain errors 5.User satisfaction is evaluated using questionnaires and interviews 6.Field observations may be used to provide contextual understanding 7.Involves recording performance of typical users doing typical tasks

Usability Testing and Experiments

Usability Testing	Experiments for Research
→ Improve products	→ Discover knowledge
→ Few participants	→ Many participants
→ Results inform design	→ Results validated statistically
→ Usually not completely replicable	→ Must be replicable
→ Conditions controlled as much as possible	→ Strongly controlled conditions
→ Procedure planned	→ Experimental design
→ Results reported to developers	→ Scientific report to scientific community

Experimental design (Controlled settings)

- Test hypothesis
- Predict the relationship between two or more variables
- Independent variable is manipulated by the researcher
- Dependent variable influenced by the independent variable
- Typical experimental designs have one or two independent variables
- Validated statistically and replicable

Types of Hypotheses

- **Null hypothesis(H0):** typically states that there is no difference between experimental treatments
- **Alternative hypothesis(H1):** a statement that is mutually exclusive with the null hypothesis
- The goal of an experiment is to find statistical evidence to reject the null hypothesis in order to support the alternative hypothesis
- H0: There is no difference in dependent variable(测量的东西) between independent variable(不同组/不同条件).
- Experimental designs:**
- 1.**Between subjects design:** Different participants, Single group of participants is allocated randomly to the experimental conditions. **Pros:**No order effects. **Cons:**Many subjects and individual difference is a problem.
- 2.**Within subjects design:** Same participants, All participants appear in both conditions. **Pros:**Few individuals, no individual differences. **Cons:** Counter-balancing needed because of ordering effects.

Experimental design lifecycle

- 1.Identify a research hypothesis
- 2.Specify the design of the study
- 3.Run a pilot study to test the design, the system, and the study instruments
- 4.Recruit participants
- 5.Run the actual data collection sessions
- 6.Analyze the data
- 7.Report the results

Experiment procedure (Data collection)

- 1.Preparation 2.Greet participants 3.Introduce the purpose of the study and the procedures
- 4.Get consent 5.Assign participants to a specific experiment condition 6.Training task(s)
- 7.Actual task(s) 8.Participants answer questionnaires (if any) 9.Debriefing session
- 10.Payment (if any)

Statistical analysis: count, average, standard deviation

F-test 用来检查两组样本的方差是否有显著差异。

T-test 用来检查两组样本的平均值是否有显著差异。

Observation

Types: 1.Direct observation: (1)In the field (2)In controlled environments 2.Indirect observation: tracking users’ activities

Direct observation in the field: Three easy-to-remember parts: The person: Who? The place: Where? The thing: What? 需要考虑的问题: Decide on how involved you will be, from passive observer to active participant? How to gain acceptance? How to handle sensitive topics, for example, culture, private spaces, and so on? How to collect the data? Ethnograph 民族志:Participation is the key. Richer and more detailed than other methods, but expensive and challenging.

Field Study (natural settings)

Seek to understand what users do naturally and how technology impacts them. Field studies are used in product design to:1.Identify opportunities for new technology 2.Determine design requirements 3.Decide how best to introduce new technology 4.Evaluate technology in use

Analytics

A variety of users’ actions can be recorded by software automatically. **Pros** of logging activity automatically: 1.It is unobtrusive provided the system’s performance is not affected 2.Large volumes of data can be logged automatically and then explored and analyzed using visualization and other tools. **Cons:** It raises ethical concerns about observing participants if this is done without their knowledge.

Web Analytics: A form of interaction logging that analyses users’ activities on website. Designers use the analysis to improve their designs. Enable designers to track the activities of users on their site.

A/B Testing

1.A large-scale experiment (thousands of participants or more). 2.Offers another way to evaluate a website, application of app running on a mobile device. 3.Often used for evaluating changes in design on social media applications. 4.Compares how two groups of users perform on two versions of a design. 5.May create ethical dilemmas if users don’t know they are part of the test.

Predictive Modeling

- 1.Provide a way of evaluating products or designs without directly involving users, less expensive than user testing. 2.Use formulas to derive various measures of user performance. 3.Usefulness limited to systems with predictable tasks, for example, voicemail systems, smartphones, and dedicated mobile devices
- Summary**
- Inspections can be used to evaluate requirements, mockups, functional prototypes, or systems
- User testing and heuristic evaluation may reveal different usability problems
- Walkthroughs are a fine-grained focused method for evaluating small parts of a product
- Analytics involves collecting data about users activity on a website or product to see which parts are used
- A/B testing is a form of large-scale experiment
- Fitts’ Law can be used to predict expert, error-free performance for clearly defined tasks with limited key presses, for example, to evaluate keypress sequences for handheld devices and the position of objects on a screen.

Part 2 Prototype 相关

Definition: A prototype is an early model or representation of a product designed to test an idea or function before committing to full production. It transforms abstract concepts into tangible experiences that can be evaluated, tested, and refined.

In design thinking, prototyping is the **bridge** between ideation and testing, where imagination meets execution.

Characteristics of Prototyping: Goal-Oriented, Iterative, Disposable & Flexible.

Why prototype?

- 1. **Evaluation and feedback** are central to interaction design 2.Stakeholders can see, hold, interact with a prototype more easily than a document or a drawing 3.Team members can communicate effectively 4.You can test out ideas for yourself 5.It encourages reflection: very important aspect of design 6.Prototypes answer questions, and support designers in choosing between alternatives

Purposes to use Prototype

1.Understanding & Ideation: Embrace

Ambiguity: Use early prototypes to ask questions, not just to provide definitive answers. **Keep Flexible:** Focus details too early can stifle divergent thinking. Stick to sketches or simple wireframes initially. **Generate Variants:** Build multiple, distinctly different versions of a feature to thoroughly explore the solution space. **Intangible -> tangible.** User-centered process/design: **build empathy** between designers and users. Empathy/user journey map.

- 2.**Communication: Tailor to the Audience:** Executives may need a high-level interactive flow, while developers need functional logic and edge cases. **Focus on Narrative:** Use the prototype to tell a story about the user journey, rather than just clicking through isolated screens. **Direct the Feedback:** Explicitly tell stakeholders what you are testing (e.g., Ignore the colors) to prevent unhelpful critiques.
- 3.**Test & Reflection:** Define clear metrics, observe user behaviour, reflect and iterate.
- Hypotheses and assumptions.** Within the design group, present and promote the idea to peers and reflect.

Achieving Prototyping Goals

To effectively leverage these three purposes, maintain these overarching principles:1.Align Purpose and Project Stage 2.Avoid the “Polish Trap” 3.Foster a Learning Culture

Fidelity in prototyping

Definition: The fidelity of a prototype refers to how it conveys the look-and-feel of the final product.

A Measurement, Not a Classification. It is a spectrum.

Why: The Cost Tension, The Need for a Vocabulary, Validating Assumptions

Advantages: Resource allocation, directing feedback, and multi-dimensional testing.

Tools for prototyping

- 1.**Storyboards:** Visualizing the Context.
- Primary Benefit:** By placing the product in a realistic environment, storyboarding builds deep empathy and highlights environmental constraints that standard wireframes miss.
- 2.**Use journey Map:**show the complete experience a user has with your product over time. It helps teams align on customer priorities and identify opportunities for improvement. Key Components:User Persona,Journey Stages, User Actions,Emotions and Thoughts, Pain Points.
- 3. **Empathy map:**建立对用户同理心, 把用户的感受想法行为和需求可视化.用户放在中间然后分几个区域:says,thinks,does,feels...
- 4. **‘Wizard-of-Oz’:**The user thinks they are interacting with a computer, but a developer is responding to output rather than the system. Usually done early in design to understand users’ expectations.

Five dimensions in prototyping

- 1.**Visual fidelity:** Refers to the devotion and granularity put into the design for visual interfaces and physical objects. It is the most direct dimension that user feels about the prototype.
- The big, obvious problem involves jumping to high visual fidelity too early in the process. Just because a designer is capable of prototyping fairly quickly at high visual fidelity does not make it the right decision.
- 2.**Interaction fidelity:** Refers to the objective degree of exactness with which real-world interactions can be reproduced. Users can click, tap, and swipe through screens and get the feel. It allowed the team to quickly and cheaply test hundreds of variations.
- 3.**Breadth fidelity:** Refers to the scope of features available.

It’s up to you to decide because the entry and exit points to your redesign may matter more than you think.

Horizontal Prototyping: A horizontal prototype models the surface layer of the entire user interface. **Primary Focus:** Evaluating the Information Architecture (IA) and overall navigation paradigm before committing to complex backend development.

Strategic Uses for Horizontal Prototyping: Testing Navigation & Categorization, Stakeholder Alignment & Buy-In, Rapid Conceptual Iteration

4.**Depth fidelity:** Refer to the degree of functionality within a single feature. Remind yourself about depth by asking **if the interactions are shallow.**

Vertical Prototyping: Acts as a vertical slice through the application’s layers from the frontend user interface down to the backend logic and the database.

Strategic Uses for Vertical Prototyping: Evaluating Complex Task Usability, Assessing Technical Feasibility, Testing Error Recovery States

5.**Content fidelity:** Refers to the degree of realism for the content.

6.**Other dimensions:** 6.Autonomy: operates alone vs requires“supervision” 7.Platform: interim vs final implementation.

Each prototyping tool has strengths and weaknesses. It is complicated (slow, expensive)

to prototype multiple dimensions at once.

Concrete Design

Definition:Considers more concrete, detailed issues of designing the interface.Iteration between concrete and conceptual design.From "What the system does" to "How the user interacts".

Menu design:Structuring navigation choices into specific formats(dropdowns, accordions, mega-menus) based on context.

Icon design: Creating visual metaphors that communicate functionality instantly, balancing style with universal recognition. Good icon design is difficult. Meaning is cultural and context sensitive.When in doubt, label it.

Screen design: How to split across screens (Task analysis as a starting point, Keep information available). Individual screen design(Draw user attention to salient point, Animation is very powerful but can be distracting, good organization helps,trade off between sparse population andovercrowding) **Information display:** Relevant information available always,different types of information imply different display,Consistency between paper display and screen data entry.

Interfaces and Research Considerations

Interface Metaphors

Why: Interface designed to be similar to a physical entity but also has own properties. Exploit user’s familiar knowledge, helping them to understand ‘the unfamiliar’

Benefits: 1.Makes learning new systems easier 2.Helps users understand the underlying conceptual model 3.Can be very innovative and enable the realm of computers and their applications to be made more accessible to a greater diversity of users

- Problems:** 1.Break conventional and cultural rules 2.Can constrain designers in the way that they conceptualize a problem space 3.Conflicts with design principles 4.Forces users to understand only the system in terms of the metaphor 5.Designers can inadvertently use bad existing designs and transfer the bad parts over
- 6.Limits designers’ imagination in coming up with new conceptual models

Interaction types

- 1.**Instructing:** Where users instruct a system and tell it what to do. The main benefit is that instructing supports quick and efficient interaction.
- 2.**Conversing:** Underlying model of having a conversation with another human. Ranges from simple voice recognition menu-driven systems to more complex ‘natural language’ dialogs.
- Pros:** Allows users, especially novices, to interact with a system in a way that is familiar to them. **Cons:** Misunderstandings can arise when the system does not know how to parse what the user says
- 3.**Manipulating:** Exploits users’ knowledge of how they move and manipulate in the physical world. Involves dragging, selecting, opening, closing and zooming actions on virtual objects

- 4.**Exploring:** Involves moving through virtual or physical environments.
- 5.**Responding:** System takes the initiative to alert user to something that it “thinks” is of interest. Automatic system response without any requests made by the user

Choosing an interaction type: Direct manipulation is good for ‘doing’ types of tasks, Issuing instructions is good for repetitive tasks, Having a conversation is good for certain services ,Hybrid conceptual models are good for supporting multiple ways of carrying out the same actions

Interaction and interface

Interaction: A description of what the user is doing when interacting with a system **Interface:** The kind of interface used to support the interaction

Interfaces

- 1.**WIMP:**Xerox Star first WIMP gave rise to GUIs: **Windows:** Sections of the screen that can be scrolled, stretched, overlapped, opened, closed, and moved around the screen using the mouse. **Icons:** Pictograms that represent applications, objects, commands, and tools that were opened when clicked on. **Menus:** Lists of options that can be scrolled through and selected. **Pointing device:** A mouse controlling the cursor as a point of entry to the windows, menus, and icons on the screen

- 2.**Menu styles**→ Flat list: Good for showing large number of options at the same time when display is small→ Drop down: Shows more options on same screen (for example, cascading)→ Pop-up: When pressed, command key for relevant options→ Contextual: Provides access to often-used commands associated with a particular item→ Collapsible: Toggles between + and – icons on a header to expand or contract its contents→ Mega: All options shown using 2D drop-down layout
- 3.**Torchscreen:**Single touchscreens are used in walk-up kiosks to detect the presence and location of a person’s touch on the display. Multi-touch surfaces support a range of more dynamic finger tip actions. They do so by registering touches at multiple locations using a grid. Supports one and two hand gestures, including tapping, zooming, stretching, flicking, dwelling, and dragging.